

FMNL2 regulates actin for ER and mitochondria distribution in oocyte meiosis

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Abstract

During mammalian oocyte meiosis, spindle migration and asymmetric cytokinesis are unique steps for the successful polar body extrusion. The asymmetry defects of oocytes will lead to the failure of fertilization and embryo implantation. In present study we reported that an actin nucleating factor formin-like 2 (FMNL2) played critical roles in the regulation of spindle migration and organelle distribution. Our results showed that FMNL2 mainly localized at the oocyte cortex and periphery of spindle. Depletion of FMNL2 led to the failure of polar body extrusion and large polar bodies in oocytes. Live-cell imaging revealed that the spindle failed to migrate to the oocyte cortex, which caused polar body formation defects, and this might be due to the decreased polymerization of cytoplasmic actin by FMNL2 depletion. Furthermore, mass spectrometry analysis indicated that FMNL2 was associated with mitochondria and endoplasmic reticulum-related proteins, and FMNL2 depletion disrupted the function and distribution of mitochondria and endoplasmic reticulum, showing with decreased mitochondrial membrane potential and the occurrence of endoplasmic reticulum stress. Microinjecting *Fmnl2*-EGFP mRNA into FMNL2-depleted oocytes significantly rescued these defects. Thus, our results indicate that FMNL2 is essential for the actin assembly, which further involves into meiotic spindle migration and ER/mitochondria functions in mouse oocytes.

eLife assessment

This study presents **useful** findings regarding the role of formin-like 2 in mouse oocyte meiosis. Some of the data are supported by **incomplete** methodological details and analyses, and several conclusions are overstated. This paper would be of interest to reproductive biologists.

Introduction

Mammalian oocyte maturation is an asymmetric division process that generates a large egg and a small polar body. This asymmetry is critical for the following fertilization and early embryo development. After germinal vesicle breakdown (GVBD), the meiotic spindle is organized at the center of the oocyte, and then it migrates to the oocyte cortex at the late metaphase I (MI). The oocytes are arrested at metaphase II (MII) after the extrusion of first polar body (1, 2). Actin filaments, as the most widely distributed cytoskeleton in cells, regulate various dynamic events during oocyte meiotic maturation (3), and two key events are the spindle migration and cortical reorganization in mammalian oocytes (1, 4, 5). Small GTPases and actin nucleation factors are shown to promote the assembly and function of actin. The actin nucleation factors are the molecules that directly promote the actin assembly: Arp2/3 complex control the assembly of branched actin, and formin family member Formin2 (FMN2) and Spire1/2 control the assembly of linear actin. These proteins are all proposed to play a role in actin-related spindle migration and cytokinesis during mammalian oocyte maturation (6–8). The cortex protein Arp2/3 complex nucleates the actin to produce a hydrodynamic force to move the spindle toward the cortex, and regulates cytokinesis during oocyte maturation (1, 8). FMN2 and Spire1/2 nucleates actin around the spindle in the cytoplasm to give the meiotic spindle an initial power for migration (7, 9).

Besides Formin2, the DRFs (diaphanous-related formins) subfamily in the formin family has been extensively studied. The DRFs family consists of mDia, Daam, FHOD and FMNLs (10). The “Formin-like” proteins (FMNLs) subfamily includes FMNL1 (FRL1), FMNL2 (FRL3), and FMNL3 (FRL2). Like other Formin family proteins, FMNLs play important roles in cell migration, cell division, and cell polarity (10, 11). While FMNL2 is widely expressed in multiple human tissues, especially in the gastrointestinal and mammary epithelia, lymphatic tissues, placenta, and reproductive tract (12). As an important actin assembly factor, FMNL2 accelerates the elongation of actin filaments branched by Arp2/3 complex (13). In invasive cells, FMNL2 is mainly localized in the leading edge of the cell, lamellipodia and filopodia tips, to improve cell migration ability by actin-based manner (13–15). FMNL2 is also involved in the maintenance of epithelial-mesenchymal transition (EMT) in human colorectal carcinoma cell (16). Besides its roles on the actin assembly, emerging evidences indicate that FMNL2 may interact with organelle dynamics. It is shown that FMNL2 is related with the Golgi apparatus, since the absence of FMNL2/3 can cause the Golgi fragmentation (17). However, till now the roles of FMNLs especially FMNL2 on oocyte meiosis are still largely unknown.

In the present study, we disturbed the FMNL2 expression and explored the roles of FMNL2 during mouse oocyte meiosis. Our results showed that FMNL2 was essential for the polar body size control and successful extrusion; and these abnormal phenotypes might be due to aberrant actin-based meiotic spindle migration. Meanwhile, we also found that FMNL2 was essential for the functions and distribution of mitochondria and endoplasmic reticulum. Therefore, this study provided the evidence for the critical roles of FMNL2-mediated actin on spindle movement and organelle dynamics in mammalian oocytes.

Materials and Methods

Antibodies and chemicals

Rabbit monoclonal anti-FMNL2 antibody, rabbit monoclonal anti-Arp2 antibody, mouse monoclonal anti-profilin1 antibody were from Santa Cruz (Santa Cruz, CA, USA). Rabbit monoclonal anti-Fascin antibody was purchased from Abcam (Cambridge, UK). Rabbit polyclonal anti-INF2 antibody was purchased from Proteintech (Proteintech, CHI, USA). Rabbit monoclonal anti- α -tubulin (11H10) antibody, rabbit monoclonal anti-Grp78 antibody, rabbit monoclonal anti-cofilin antibody and rabbit monoclonal anti-Chop antibody were from Cell Signaling Technology (Beverly, MA, USA). Mouse monoclonal anti- α -tubulin-FITC antibody was from Sigma-Aldrich Corp. (St. Louis, MO, USA). FITC-conjugated goat anti-rabbit IgG were from Zhongshan Golden Bridge Biotechnology (Beijing). ER-Tracker Red, Mito-Tracker Green and enhanced mitochondrial membrane potential assay Kit were from Beyotime Biotechnology (Shanghai). All other chemicals and reagents were from Sigma-Aldrich Corp., unless otherwise stated.

Ethics statement and oocyte culture

We followed the guidelines of Animal Research Institute Committee of Nanjing Agricultural University to conduct the operations. The animal facility had license authorized by the experimental animal committee of Jiangsu Province (SYXK-Su-20170007). The Female Institute of Cancer Research (ICR) mice, aged 4–6 weeks, were kept in a room with a regulated temperature of 22 °C and provided with a standard diet. Fully developed germinal vesicle stage (GV) oocytes were retrieved from the ovaries of mice, and then cultured in M16 medium with paraffin oil at 37 °C and in the presence of 5% CO₂ for in vitro maturation. At specific intervals, the oocytes were collected for various tests and analyses. The oocytes were placed at 37°C with an atmosphere of 5% CO₂, and cultured to different time points for immunostaining, microinjection and western blot.

Plasmid construct and in *vitro* transcription

Template RNA was generated from mouse ovaries with RNA Isolation Kit (Thermo fisher), then we reversed transcription of these RNA to create cDNA by a PrimeScript 1st strand cDNA synthesis kit (Takara, Japan). FMNL2-EGFP vector was generated by Wuhan GeneCreate Biological Engineering Co, Ltd. mRNA was synthesized from linearized plasmid using HiScribe T7 high yield RNA synthesis kit (NEB), then capped with m7G (5') ppp (5') G (NEB) and tailed with a poly(A) polymerase tailing kit (Epicentre) and purified with RNA clean & concentrator-25 kit (Zymo Research).

Microinjection of FMNL2 siRNA and mRNA

The FMNL2 siRNA working solution was dissolved in RNase-free water, achieving a concentration of 20 μ M. For FMNL2 knockdown (KD), three individual siRNA strands were precisely mixed and subjected to centrifugation to obtain the supernatant, approximately 5-10 pL of supernatant was microinjected into the cytoplasm of GV stage oocytes. In contrast, an equal volume of a negative control solution was microinjected into the cytoplasm of oocytes in the control group. FMNL2 siRNA: 5'-GCU GAA UGC UAU GAA CCU ATT-3', 5'-GCC AUU GAU CUU UCU UCA ATT-3', 5'-GGA AUU AAG AAG GCG ACA ATT-3'; Negative control siRNA: 5'-UUC UCC GAA CGU GUC ACG UTT-3'. Then, the oocytes were arrested in the GV stage for 18h in M16 medium with 5 μ M milrinone. This was done to optimize the effectiveness of the siRNA and aid in the depletion of FMNL2. For the rescue experiment, 5–10 pL of 200ng/ μ L FMNL2-EGFP mRNA was injected into the GV oocytes 18h after FMNL2 siRNA injection. Following that, the GV stage oocytes were cultured in M16 medium with 5 μ M milrinone for 4h. Then, the oocytes being cultured in fresh M16 medium for subsequent experiments.

Immunofluorescent staining and confocal microscopy

Oocytes were fixed in 4% paraformaldehyde (in PBS) for 30 min and permeabilized with 0.5% Triton X-100 in PBS for 20 min then blocked in blocking buffer (1% BSA-supplemented PBS) at room for 1 h. For FMNL2 staining, the oocytes at various stages (GV, GVBD, MI, and MII) were incubated with Rabbit monoclonal anti-FMNL2 antibody (1:100) at 4 °C overnight, then oocytes were washed by wash buffer (0.1% Tween 20 and 0.01% Triton X-100 in PBS) for 3 times (5 min each time). Next the oocytes were labeled with secondary antibody coupled to FITC-conjugated goat anti-rabbit IgG (1:100) at room temperature for 1 h. For α -tubulin staining, MI stage oocytes were incubated with anti- α -tubulin-FITC antibody (1:200). For actin staining, GV and MI stage oocytes were incubated with Phalloidin-TRITC at room temperature for 2 h. Then the oocytes were washed as the same way. Finally, oocytes were incubated with Hoechst 33342 at room temperature for 10-20 min. After staining, samples were mounted on glass slides and observed with a confocal laser-scanning microscope (Zeiss LSM 800 META, Germany).

ER and Mito-tracker staining

To study ER and mitochondria distribution during mouse oocyte meiosis, MI stage oocytes were incubated with ER-Tracker Red (1:3000) or 200 nM Mito-tracker green (Red) in M16 medium for 30 min at 37°C and 5% CO₂. Then the oocytes were washed three times with M2 medium, finally the samples were examined with confocal microscopy. During the MI stage of oocyte development, both the ER and mitochondria evenly distributed in the cytoplasm and accumulated at the spindle periphery in MI stage. This clustering phenomenon is considered as the normal localization pattern of these organelles. On the other hand, when the ER and mitochondria cluster randomly within the cytoplasmic region, this localization is defined as organelle abnormal localization pattern.

JC-1 detection

The enhanced mitochondrial membrane potential assay Kit was employed to analyze the mitochondrial membrane potential of oocytes. The MI stage oocytes were transferred from the M16 medium to JC-1 for 30 min at 37°C and 5% CO₂. Following three washes with M2 medium, the oocytes were examined using a fluorescent microscope (OLYMPUS IX71, Japan) for the presence of a fluorescent signal.

Time lapse microscopy

To image the dynamic changes that occurred during oocyte maturation, oocytes were cultured in M16 medium, then transferred to the Leica SD AF confocal imaging system equipped with 37 °C incubator and 5% CO₂ supply (H301-K-FRAME). The spindle in oocytes was labeled by α -tubulin-EGFP.

Immunoprecipitation

4-6 ovaries were put into RIPA Lysis Buffer contained phosphatase inhibitor cocktail (100×) (Kangwei Biotechnology, China), and were completely cleaved on ice block. We collected supernatant after centrifugation (13200 rps, 20 min) and then took out 50 μ l as input sample at 4 °C. The rest of the supernatant was incubated with primary antibody (FMNL2 or INF2 antibody) overnight at 4 °C. 30 μ l conjugated beads (washed five times in PBS) were added to the supernatant/antibody mixture and incubated at 4 °C for 4-6 h, after three times wash by immune complexes, the samples were then released from the beads by mixing in 2× SDS loading buffer for 10 min at 30 °C.

Western blot analysis

Approximate 100-150 GV or MI stage mouse oocytes were placed in Laemmli sample buffer and heated at 85°C for 7-10 min. Proteins were separated by SDS-PAGE at 165V for 70-80 min and then electrophoretically transferred to polyvinylidene fluoride (PVDF) membranes (Millipore, Billerica, MA, USA) at 20 V for 1 hour. After transfer, the membranes were then blocked with TBST (TBS containing 0.1% Tween 20) containing 5% non-fat milk at room temperature for 90 min. After blocking, the membranes were incubated with rabbit monoclonal anti-FMNL2 antibody (1:500), rabbit monoclonal anti-Arp2 antibody (1:500), mouse monoclonal anti-profilin1 antibody (1:500), rabbit monoclonal anti-Fascin antibody (1:5000), rabbit polyclonal anti-INF2 antibody (1:500), rabbit monoclonal anti-Grp78 antibody (1:1000), rabbit monoclonal anti-cofilin antibody (1:2000), rabbit monoclonal anti-CHOP antibody (1:1000), or rabbit monoclonal anti-tubulin antibody (1:2000) at 4 °C overnight. After washing 5 times in TBST (5 min each), membranes were incubated for 1h at room temperature with HRP-conjugated Pierce Goat anti-Rabbit IgG (1:5000) or HRP-conjugated Pierce Goat anti-mouse IgG (1:5000). After washing for 5 times, the membranes were visualized using chemiluminescence reagent (Millipore, Billerica, MA). Every experiment repeated at least 3 times with different samples.

Fluorescence Intensity Analysis

Immunofluorescence experiments were conducted simultaneously and with consistent parameters in both the control and treatment groups. The images were consistently captured using identical confocal microscope settings. Subsequently, the average intensity of fluorescence per unit area in the designated region of interest was quantified following the fluorescence staining. The acquired fluorescence data was analyzed employing ZEN 2011 and ImageJ software.

Statistical analysis

All statistical analyses were performed using GraphPad Prism7.00 software (GraphPad, CA, USA), employing the t-test to assess the statistical significance between the control and treatment groups. The results were represented as the mean $\pm$ standard error of the mean (SEM). Statistical significance was defined as a P-value < 0.05 , denoted as *, ** for $P < 0.01$, ***, and **** for $P < 0.001$ and $P < 0.0001$, respectively. Every experiment was conducted with a minimum of three biological replicates.

Results

Expression and subcellular localization of FMNL2 during mouse oocyte maturation

We first examined FMNL2 expression in mouse oocytes at different stages. The results indicated that FMNL2 all expressed in GV, MI and MII stages during mouse oocyte maturation (GV, 1; MI, 0.82 ± 0.07 ; MII, 0.61 ± 0.10 , **Figure 1A**). Next, we performed FMNL2-EGFP mRNA microinjection to examine the localization of FMNL2. As shown in **Figure 1B**, FMNL2 accumulated at the oocyte cortex during the GV, GVBD MI and MII stages. Besides, FMNL2 also localized at the spindle periphery during MI stages. The FMNL2 antibody staining results also confirmed this localization pattern. In addition, we co-stained FMNL2 antibody with F-actin, and the results revealed that both FMNL2 and F-actin are localized in the cortex region of oocytes (**Figure 1C**). The FMNL2 localization pattern indicated that FMNL2 might interact with actin dynamics during oocyte meiosis.

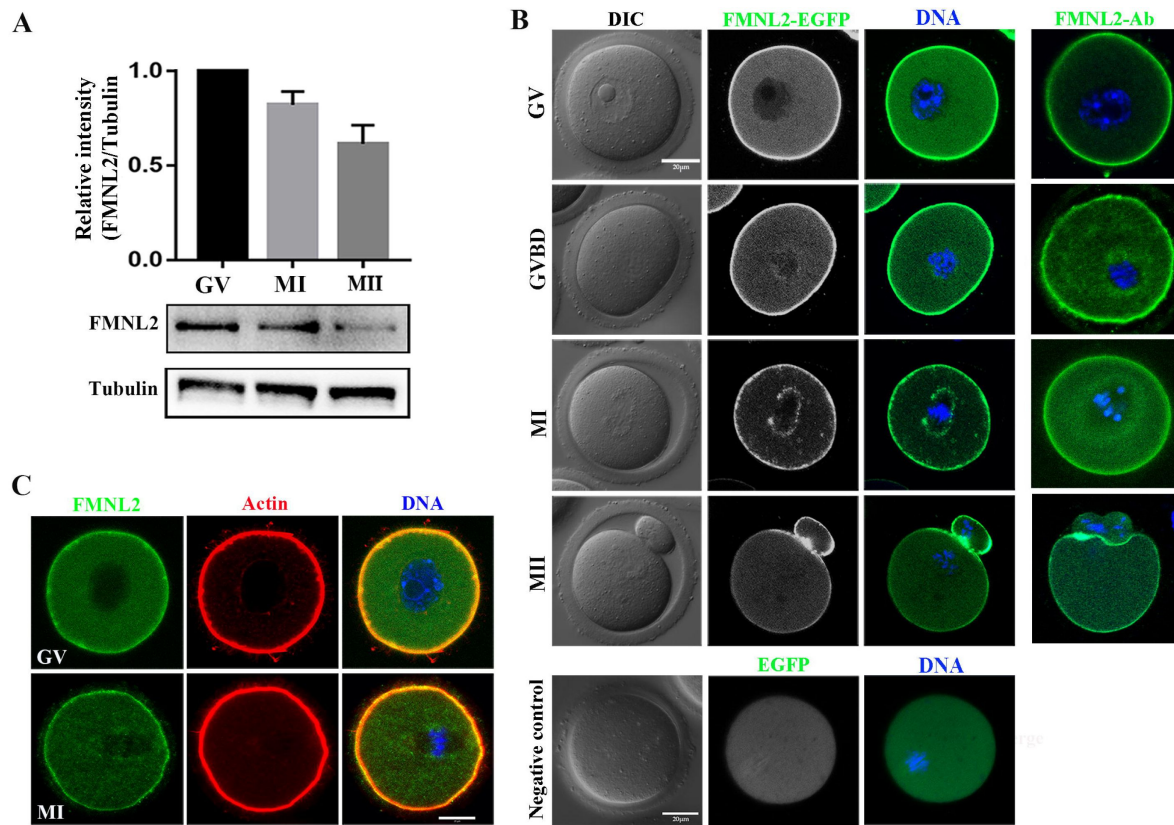


Figure 1.

Expression and subcellular localization of FMNL2 during mouse oocyte meiosis.

(A) Western blotting results of FMNL2 protein expression at different stages. FMNL2 expressed at the GV, MI, and MII stages. (B) Subcellular localization of FMNL2-EGFP and FMNL2 antibody during mouse oocyte meiosis. FMNL2 was enriched at the cortex (GV, GVBD, MI and MII stage) and spindle periphery (MI stage). Green, FMNL2-EGFP; blue, DNA. Negative control: Green, EGFP; blue, DNA. Bar =20 μ m. (C) Co-staining of oocytes for FMNL2 and actin. FMNL2 and actin both localization in cortex. Green, FMNL2-antibody; red, actin; blue, DNA. Bar =20 μ m.

FMNL2 is essential for polar body extrusion and asymmetric division in mouse oocytes

To investigate the functional roles of FMNL2 in mouse oocytes, we employed FMNL2 siRNA microinjection to knockdown FMNL2 protein expression. A significant decrease of FMNL2 protein level was shown in FMNL2-KD oocytes compared to control group by western blot (1 vs. 0.48 ± 0.08 , $P < 0.01$, **Figure 2A**). We then examined the maturation of oocytes, and the results indicated that knockdown of FMNL2 had an impact on the extrusion of the first polar body. Moreover, a significant proportion of oocytes exhibited larger polar bodies upon extrusion (**Figure 2B**). Based on the size of the extruded polar bodies, those with a diameter exceeding one-third of the oocyte's diameter were categorized as large polar bodies. Consequently, we proceeded to calculate the rates of polar body extrusion and the generation of large polar bodies in the oocytes. The quantitative results also confirmed this phenotype (rate of polar body extrusion: $74.26 \pm 1.44\%$, $n = 439$ vs. $59.5 \pm 2.82\%$, $n = 398$, $P < 0.001$, **Figure 2C**; rate of large polar bodies: $19.05 \pm 1.97\%$, $n = 311$ vs. $37.16 \pm 1.87\%$, $n = 257$, $P < 0.0001$, **Figure 2D**). In addition, live-cell imaging was used to determine the dynamic changes that occurred during oocyte maturation, and the results showed that the oocytes either failed to undergo cytokinesis or divided from the central axis of the oocytes and formed big polar bodies (**Figure 2E**). To further confirm the phenotype, we performed FMNL2 rescue experiments by expressing exogenous Fmnl2 mRNA in FMNL2-depleted oocytes (**Figure 2F**), we found that exogenous Fmnl2 mRNA expression rescued first polar body extrusion and large polar body defects (**Figure 2G**). The quantitative results also confirmed this phenotype (rate of polar body extrusion: $48.34 \pm 4.2\%$, $n = 355$ vs. $62.62 \pm 3.6\%$, $n = 377$, $P < 0.01$, **Figure 2H**; rate of large polar bodies: $30.93 \pm 2\%$, $n = 193$ vs. $9.58 \pm 2.4\%$, $n = 203$, $P < 0.01$, **Figure 2I**). It is known that knockdown of FMNL3 leads to inhibition of oocyte maturation. To investigate whether FMNL2 exhibits an additive effect with FMNL3 in terms of functionality, we simultaneously knocked down both FMNL2 and FMNL3. The results demonstrated that simultaneous knockdown of the two FMNL proteins, compared to the control group, resulted in a decrease in oocyte maturation rate. However, when compared to the single knockdown of FMNL2, the double knockdown of FMNL2 and FMNL3 did not cause more severe defects in polar body extrusion. (polar body extrusion, Control: $70.97 \pm 1.23\%$, $n=261$ vs FMNL2+3-KD: $60.42 \pm 2.99\%$, $n=198$, $P < 0.05$, **Figure 2J**). Large polar body, Control: $10.85 \pm 0.97\%$, $n=172$ vs FMNL2+3-KD: $32.90 \pm 1.88\%$, $n=118$, $P < 0.001$, **Figure 2K**). These results suggested that FMNL2 played critical roles for the polar body extrusion and asymmetric division during mouse oocyte maturation.

FMNL2 regulates spindle migration during mouse oocyte maturation

To investigate the causes for polar body extrusion defects, we examined the spindle migration by time-lapse microscopy during oocyte meiosis. As shown in **Figure 3A**, in the control oocyte, the meiotic spindle formed in the center of the oocyte after culture 8 h and moved to the oocyte cortex at 9.5h; and the polar body was extruded at 11-12h, with a spindle formed near the cortex at MII stage. However, in FMNL2-KD oocytes, two phenotypes were observed: 1) the meiotic spindle remained in the center of the oocyte until 10 h, and then the oocytes initiated the cytokinesis at 10.5 h but failed to extrude the polar body; 2) some oocytes with arrested spindles initiated the cytokinesis but extruded a big polar body (**Figure 3A**). This indicated the failure of spindle migration after FMNL2 depletion. We analyzed the rate of cortex-localized spindle in oocytes by cultured for 9.5 h, and the result showed that the rate of migrated spindles in control oocytes was significantly higher than that in FMNL2-KD oocytes ($59.94 \pm 3.42\%$, $n = 78$ vs. $38.97 \pm 6.34\%$, $n = 64$, $P < 0.05$, **Figure 3B**). We also performed FMNL2 rescue experiments. Supplementing with

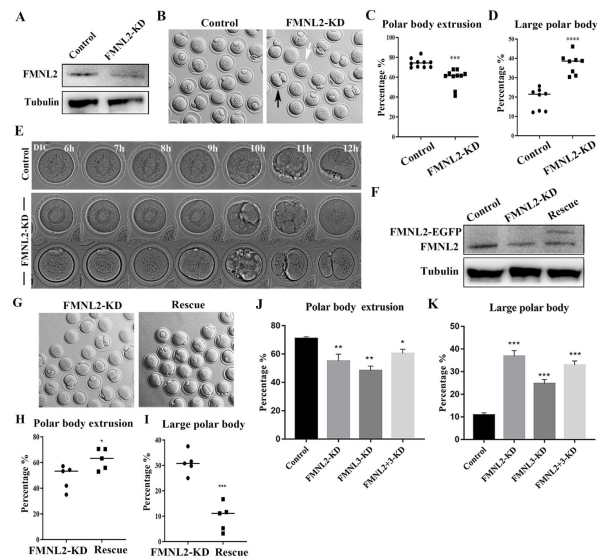


Figure 2.

Knockdown of FMNL2 affects first polar body extrusion and asymmetric division.

(A) Western blot analysis for FMNL2 expression in the FMNL2-KD group and control group. Relative intensity of FMNL2 and tubulin was assessed by densitometry. (B) DIC images of control oocytes and FMNL2-KD oocytes after 12 h culture. FMNL2-KD caused large polar bodies (black arrows) and some oocytes failed to extrude the polar bodies (white arrows). (C) Rate of polar body extrusion after 12 h culture of the control group and FMNL2-KD group. (D) Rate of large polar body extrusion after 12 h culture in the control group and FMNL2-KD group. (E) Time-lapse microscopy showed that polar body extrusion failed after FMNL2-KD. Bar = 10 μ m. (F) Western blot analysis for FMNL2 expression in the control group, FMNL2-KD group and rescue group. Relative intensity of FMNL2 and tubulin was assessed by densitometry. (G) DIC images of FMNL2-KD oocytes and rescue oocytes after 12 h culture. (H) Rate of polar body extrusion after 12 h culture of the FMNL2-KD group and rescue group. (I) Rate of large polar body extrusion after 12 h culture in the FMNL2-KD group and rescue group. (J) Rate of polar body extrusion after 12 h culture of the control group and FMNL2+3-KD group. (K) Rate of large polar body extrusion after 12 h culture in the control group and FMNL2+3-KD group. The data are presented as mean $\pm$ SEM from at least three independent experiments. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

exogenous *Fmnl2* rescued the spindle migration defects compared with the FMNL2-depletion group ($40.27 \pm 3.19\%$, $n = 181$ vs. $57.01 \pm 2.72\%$, $n = 57$, $P < 0.01$, **Figure 3C**). These results suggested that FMNL2 might be involved in spindle migration in mouse oocytes.

FMNL2 promotes cytoplasmic actin assembly during mouse oocyte maturation

As FMNL2 is a key actin assembly factor, we further investigated actin assembly after deleting FMNL2 in mouse oocytes. Surprisingly there was no significant difference for the signals of cortex actin was observed between control oocytes and FMNL2-KD oocytes, which was confirmed by the fluorescence intensity analysis (30.88 ± 1.10 , $n = 28$ vs. 30.58 ± 1.12 , $n = 28$, $P > 0.05$, **Figure 4A, 4B**). However, we found a significant decrease of cytoplasmic actin signals in the FMNL2-KD oocytes, and the statistical analysis for the cytoplasmic actin fluorescent signals also confirmed our findings (58.25 ± 2.05 , $n = 26$ vs. 37.92 ± 2.02 , $n = 24$, $P < 0.0001$, **Figure 4C, 4D**). Moreover, the rescue experiments showed that exogenous FMNL2 rescued the decrease of cytoplasmic actin filaments compared with the FMNL2-depletion group (37.98 ± 1.98 , $n = 16$ vs. 54.72 ± 2.88 , $n = 15$, $P < 0.0001$, **Figure 4E, 4F**). We next explored how FMNL2 regulates cytoplasmic actin assembly in oocytes. By mass spectrometry analysis we found there were several actin-related potential candidates which might be related with FMNL2 (**Figure 4G**). Co-immunoprecipitation results showed that FMNL2 precipitated Arp2 and Formin2 but not Profilin and fascin (**Figure 4H**). To further verify the correlation between FMNL2 and Arp2 and Formin2, we then examined Arp2 and Formin2 protein expression after FMNL2 knockdown. The results showed Arp2 protein expression increased significantly after FMNL2 knockdown (1 vs. 1.56 ± 0.07 , $P < 0.001$, **Figure 4I**) but Formin2 decreased after FMNL2 knockdown (1 vs. 0.62 ± 0.04 , $P < 0.001$, **Figure 4J**). Exogenous FMNL2 rescued these alterations compared with that in the FMNL2-KD group (Arp2 protein expression: 1 vs. 0.65 ± 0.06 , $P < 0.01$, **Figure 4I**; Formin2 protein expression: 1 vs. 1.24 ± 0.05 , $P < 0.01$, **Figure 4J**). These results indicated that FMNL2 may be associated with Formin2 and Arp2 for actin assembly in mouse oocytes.

FMNL2 regulates endoplasmic reticulum distribution during mouse oocyte maturation

The mass spectrometry analysis data indicated that several ER-related potential candidates which might be related with FMNL2 (**Figure 5A**), while INF2, a typical protein which mediates actin polymerization at ER showed high confidence level. We then examined the relationship between FMNL2 and INF2, and the co-immunoprecipitation results showed that FMNL2 precipitated INF2 and INF2 also precipitated FMNL2 (**Figure 5B**), indicating that FMNL2 interacted with INF2 in mouse oocytes. We then examined the ER distribution in FMNL2-KD oocytes. As shown in **Figure 5C**, in control oocytes the ER evenly distributed in the cytoplasm and accumulated at the spindle periphery in MI stage; however, ER agglomerated in cytoplasm in FMNL2-KD oocytes (**Figure 5C**). The statistical analysis showed that the abnormal distribution of ER increased significantly in the FMNL2-KD group (28.91 ± 5.62 , $n = 27$ vs. 59.64 ± 6.95 , $n = 28$, $P < 0.05$, **Figure 5D**). The localization pattern of ER indicated its functions might be disturbed. In FMNL2-KD oocytes, we found the expressions of ER-stress related proteins Grp78 and Chop were significantly increased (Grp78: 1 vs. 1.42 ± 0.12 , $P < 0.05$; Chop: 1 vs. 1.53 ± 0.16 , $P < 0.05$, **Figure 5E**), indicating the occurrence of ER stress. We also performed FMNL2 rescue experiments. Supplementing with exogenous *Fmnl2* rescued the ER distribution defects caused by FMNL2 knockdown (**Figure 5F**), which was supported by the statistical analysis showing that the abnormal distribution rate of ER decreased significantly in the rescue group (52.04 ± 5.29 , $n = 70$ vs. 34.91 ± 3.37 , $n = 78$, $P < 0.05$, **Figure 5G**). Moreover, Grp78 protein expression decreased in the rescue group (1 vs. 0.78 ± 0.05 , $P < 0.01$, **Figure 5H**). These results indicated that the depletion of FMNL2 affected ER distribution and caused ER stress in mouse oocytes.

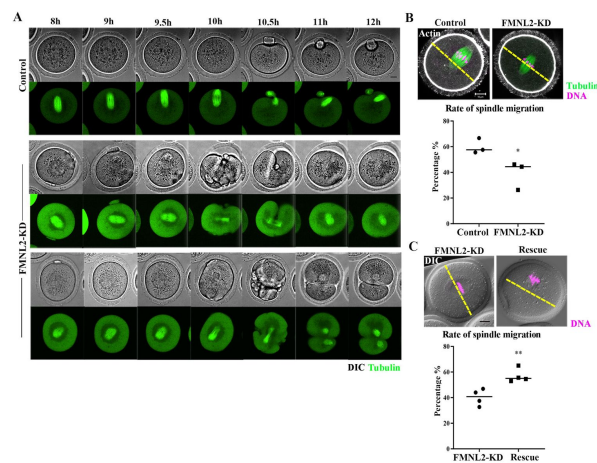


Figure 3.

Knockdown of FMNL2 disrupts spindle localization during mouse oocyte meiosis.

(A) Time-lapse microscopy showed that spindle migration failed after FMNL2-KD. Green, tubulin-EGFP. Bar = 10 μ m. **(B)** Representative images and the proportion of spindle migration after 9.5 h of culture in the control group and FMNL2-KD oocyte group. White, actin; green, tubulin; magenta, DNA. Bar = 10 μ m. **(C)** Representative images and the proportion of spindle migration after 9.5 h of culture in the FMNL2-KD group and rescue oocyte group. magenta, DNA. Bar = 10 μ m. The data are presented as mean $\pm$ SEM from at least three independent experiments. * $P < 0.05$, ** $P < 0.01$.

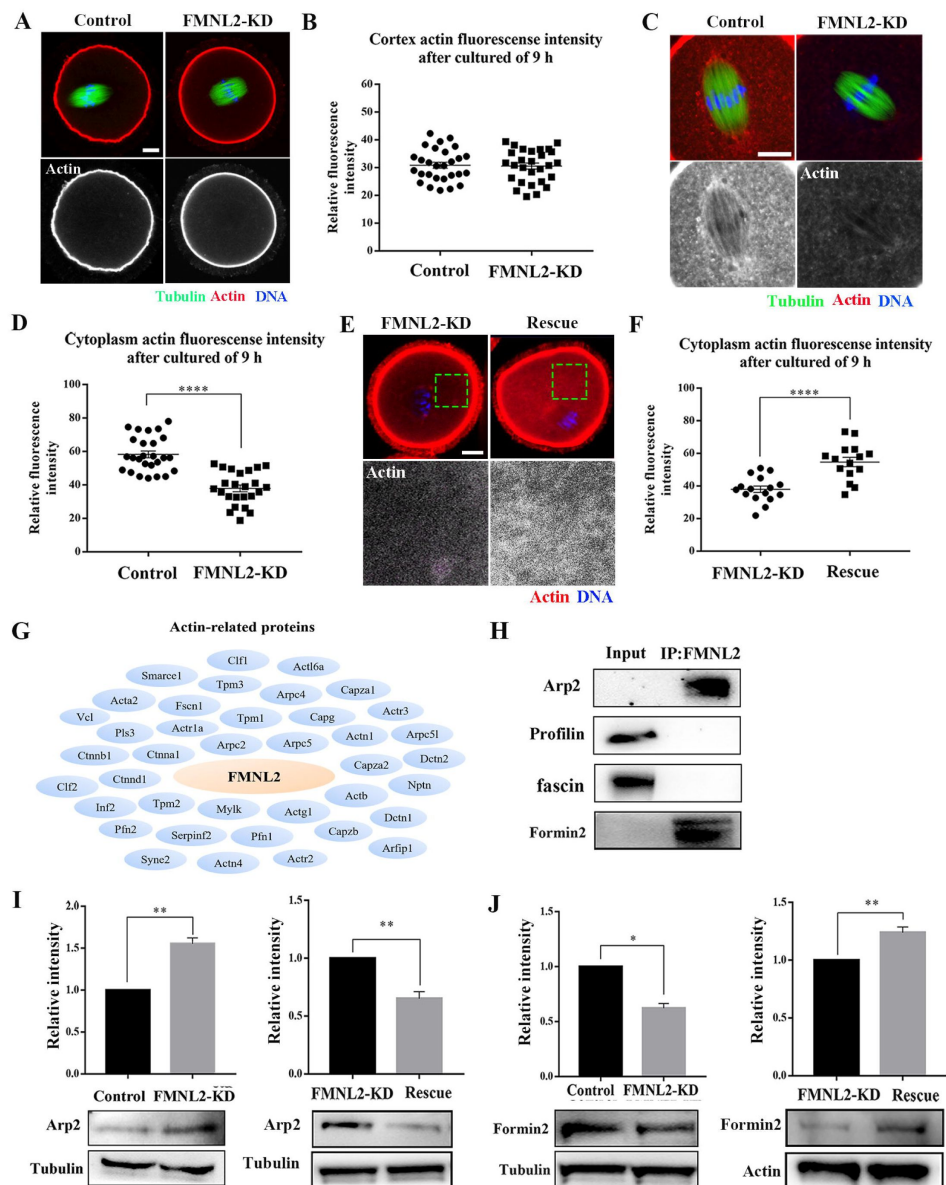


Figure 4.

Knockdown of FMNL2 disrupts actin assembly during mouse oocyte meiosis.

(A, B) Representative images of actin distribution at the oocyte cortex and the fluorescent intensities in the control group and FMNL2-KD group ($P > 0.1$). White, actin; green, tubulin; magenta, DNA. Bar = 10 μm . (C, D) Representative images of actin distribution in the oocyte cytoplasm and the fluorescent intensities in the control group and FMNL2-KD group. White, actin; green, tubulin; magenta, DNA. Bar = 10 μm . (E, F) Representative images of actin distribution in the oocyte cytoplasm and the fluorescent intensities in the FMNL2-KD group and rescue group. White, actin; magenta, DNA. Bar = 10 μm . (G) Mass spectrometry results showed that FMNL2 was related to many actin-related proteins. (H) Co-IP results showed that FMNL2 was correlated with Arp and Formin2 but not with Profilin and Fascin. (I) Arp2 protein expression significantly increased in the FMNL2-KD oocytes compared with the control oocytes. Arp2 protein expression significantly decreased in the rescue oocytes compared with the FMNL2-KD oocytes. (J) Formin2 protein expression significantly decreased in the FMNL2-KD oocytes compared with the control oocytes. Formin2 protein expression significantly increased in the rescue oocytes compared with the FMNL2-KD oocytes. The data are presented as mean $\pm$ SEM from at least three independent experiments. * $P < 0.05$, ** $P < 0.01$, **** $P < 0.0001$.

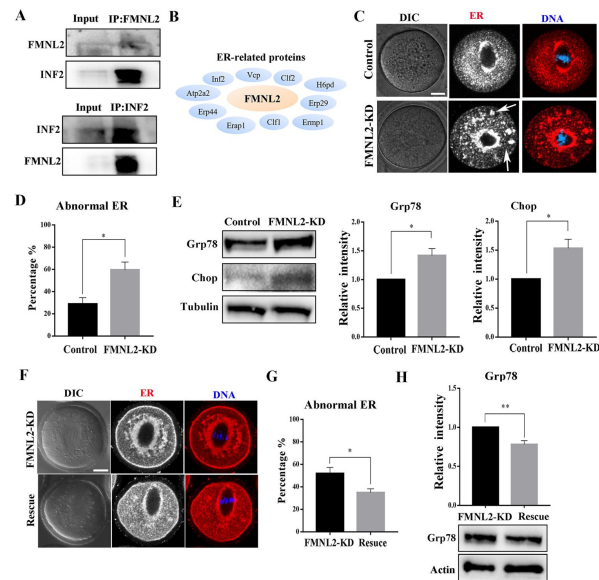


Figure 5.

FMNL2 regulates endoplasmic reticulum distribution during mouse oocytes maturation.

(A) Co-IP results showed that FMNL2 was correlated with INF2. **(B)** Mass spectrometry results showed that FMNL2 was associated with ER-related proteins. **(C)** Representative images of ER distribution in the oocyte cytoplasm in the control group and FMNL2-KD group. In FMNL2-KD oocytes, ER agglomerated in cytoplasm (white arrow). Red, ER; Blue, DNA. Bar = 10 μ m. **(D)** Abnormal distribution of ER significantly increased in the FMNL2-KD oocytes compared with the control oocytes. **(E)** Grp78 and Chop protein expression significantly increased in the FMNL2-KD oocytes compared with the control oocytes. The band intensity analysis also confirmed this finding. **(F)** Representative images of ER distribution in the oocyte cytoplasm in the FMNL2-KD group and rescue group. Red, ER; Blue, DNA. Bar = 10 μ m. **(G)** Abnormal distribution of ER significantly decreased in the rescue oocytes compared with the FMNL2-KD oocytes. **(H)** Grp78 protein expression significantly decreased in the rescue oocytes compared with the FMNL2-KD oocytes. The band intensity analysis also confirmed this finding. The data are presented as mean $\pm$ SEM from at least three independent experiments. * $P < 0.05$, ** $P < 0.01$.

FMNL2 regulates mitochondrial distribution during mouse oocyte maturation

As INF2 is also related to the mitochondrial connection of ER, we further screened up the mass spectrometry analysis data and we found many mitochondria-related potential candidates which might be related with FMNL2 (**Figure 6A** [↗](#)). Therefore, we further examined the distribution of mitochondria in FMNL2-KD oocytes. In control oocytes, the mitochondria evenly distributed in the cytoplasm and accumulated at the spindle periphery in MI stage; however, in FMNL2-KD oocytes, mitochondria presented clumped aggregation distribution in cytoplasm (**Figure 6B** [↗](#)). We counted the number of clumps and found that the uniform distribution of mitochondria decreased significantly in the FMNL2-KD group (59.66 ± 8.48 , $n = 31$ vs. 20.83 ± 4.17 , $n = 32$, $P < 0.05$, **Figure 6C** [↗](#)). A large number of FMNL2-KD oocytes agglomerated into one to three clumps (22.73 ± 4.27 , $n = 31$ vs. 42.50 ± 1.25 , $n = 32$, $P < 0.05$, **Figure 6C** [↗](#)). Supplementing with exogenous Fmnl2 rescued the mitochondria distribution (**Figure 6D** [↗](#)), the statistical analysis showed that the uniform distribution of mitochondria increased significantly in the rescue group (36.49 ± 3.97 , $n = 53$ vs. 53.90 ± 2.09 , $n = 79$, $P < 0.05$, **Figure 6E** [↗](#)). We also examined mitochondrial membrane potential, and the results showed that FMNL2 depletion caused the alterations of mitochondrial membrane potential (MMP) by JC-1 staining. The fluorescence intensity of JC-1 red channel was decreased compared with the control group (**Figure 6F** [↗](#)). We also calculated the ratio for red/green fluorescence intensity, and the results also confirmed this (control group: 0.40 vs. FMNL2-KD: 0.21 ± 0.01 , $P < 0.01$) (**Figure 6G** [↗](#)). Cofilin is an important factor of actin assembly and regulates mitochondrial function. We also examined cofilin protein expression after FMNL2 knockdown. The results showed cofilin protein expression decreased significantly after FMNL2 knockdown (1 vs. 0.81 ± 0.03 , $P < 0.01$, **Figure 6H** [↗](#)). These results indicated that FMNL2 regulated mitochondria distribution and function during mouse oocyte maturation.

Discussion

In this study, we explored the functions of FMNL2 during mouse oocyte meiosis. Our results indicated that FMNL2 regulated actin-based spindle migration for asymmetric cell division of oocytes, and more importantly FMNL2 was critical for maintaining the distribution of the ER and mitochondria, which set up a link for actin-related spindle migration and organelle dynamics in oocytes.

As a subfamily of Formin family, FMNLs play an important role in regulating actin filaments ([18](#) [↗](#)), while FMNL2 is most widely expressed in variety of cell models among the members of FMNLs. In this study, we showed that FMNL2 expressed in mouse oocytes and it mainly accumulated at the oocyte cortex and spindle periphery, which was similar with the actin distribution pattern in oocytes. This specific localization is also similar to FMN2, a well-studied factor in the formin family for spindle migration during oocyte meiosis ([7](#) [↗](#), [19](#) [↗](#)). In addition, another FMNLs family member, FMNL1 is also localized at the cortex and is essential for actin polymerization and spindle assembly during oocyte meiosis ([20](#) [↗](#)). Based on the localization pattern of FMNL2, we speculated that the functions of FMNL2 might be also involved in actin-related process during mouse oocyte meiosis.

To confirm our hypothesis, we depleted FMNL2 protein expression and we found that absence of FMNL2 caused the aberrant first polar body extrusion. The oocytes either failed to form the polar body or extruded large polar bodies. These phenotypes caused by FMNL2 depletion are similar to the other actin-related proteins during oocyte maturation such as Arp2/3 complex ([8](#) [↗](#), [21](#) [↗](#)) and FMN2 ([22](#) [↗](#), [23](#) [↗](#)). We next examined the actin distribution in oocytes since it is reported that FMNL2 promotes actin filament assembly in many models. FMNL2 is required for cell-cell

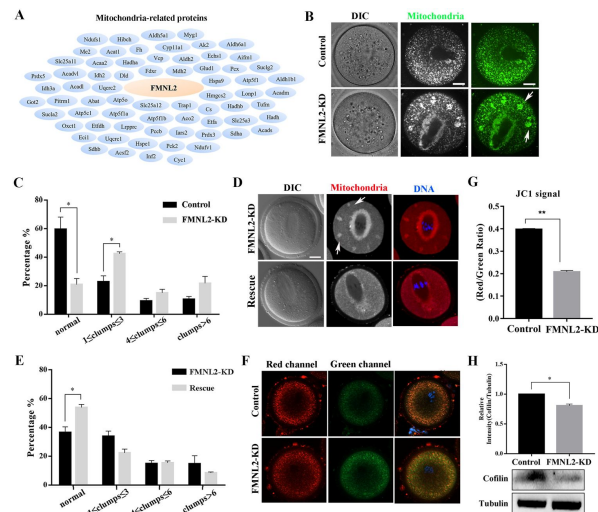


Figure 6.

FMNL2 regulates mitochondrial distribution during mouse oocytes maturation.

(A) Mass spectrometry results showed that FMNL2 was related to many mitochondria-related proteins. (B) Representative images of mitochondrial distribution in the oocyte cytoplasm in the control group and FMNL2-KD group. In FMNL2-KD oocytes, mitochondrial agglomerated in cytoplasm (white arrow). Green, Mito. Bar = 20 μ m. (C) Abnormal distribution of mitochondrial significantly increased in the FMNL2-KD oocytes compared with the control oocytes. (D) Representative images of mitochondrial distribution in the oocyte cytoplasm in the FMNL2-KD group and rescue group. In FMNL2-KD oocytes, mitochondrial agglomerated in cytoplasm (white arrow). Red, Mito; Blue, DNA. Bar = 20 μ m. (E) Abnormal distribution of mitochondrial significantly decreased in the rescue oocytes compared with the FMNL2-KD oocytes. (F) The typical picture for JC1 green channel and red channel after FMNL2-KD. (G) The JC1 signal (red/green ratio) after FMNL2-KD compare with the control group, the JC-1 red/green fluorescence ratio was significantly reduced in FMNL2-KD groups. Bar = 20 μ m. (H) cofilin protein expression significantly decreased in the FMNL2-KD oocytes compared with the control oocytes. The band intensity analysis also confirmed this finding. The data are presented as mean $\pm$ SEM from at least three independent experiments. * $P < 0.05$, ** $P < 0.01$.

adhesion formation by regulating the actin assembly (24), and FMNL2 could directly drives actin elongation (15). In CRC cells, cortactin bind to FMNL2 to active the actin polymerization, and FMNL2 is important for invadopodia formation and functions (25). Our results showed that the FMNL2 depletion caused significantly decrease in cytoplasmic actin, indicating the conserved roles of FMNL2 on actin assembly in mammalian oocyte model. Other Formin family proteins such as Daam1, FHOD1, and Formin-homology family protein mDia1 are also reported to affect oocyte meiosis by regulating actin polymerization (26–28).

We then tried to explore how FMNL2 involves into the actin assembly in oocytes. Mass spectrometry analysis data indicated that FMNL2 associated with several actin-related proteins, and we found that a potential association between FMNL2 and Arp2/Formin2. This could be confirmed by the altered expression of these two molecules after FMNL2 depletion. Therefore, we speculated FMNL2 could regulate cytoplasmic actin assembly in oocytes through the association with Formin2 since it is reported to be an important protein for cytoplasmic actin assembly in oocytes (22). Interestingly, our results showed that unlike the reduction of cytoplasmic actin, cortex actin was not affected by the absence of FMNL2. We speculate that FMNL2 and Arp2/3 both contribute to the cortex actin dynamics, when FMNL2 decreases, ARP2 increases to compensate for this, which maintains the cortex actin level. As an actin nucleator Arp2/3 complex localizes at the cortex and is essential for actin polymerization during oocyte meiosis (8, 29). These results suggested that FMNL2 might be involved in cytokinesis and asymmetric division by regulating actin assembly during mouse oocyte maturation.

The spindle migration is a key step in ensuring the asymmetric division for oocytes (30). In mitosis, spindle position is decided by cortical actin and astral microtubules; in contrast, spindle migration is mainly mediated by actin filaments during oocyte meiosis (30, 31). Due to the effects of FMNL2 on asymmetric division and cytoplasmic actin, we analyzed the spindle positioning at late MI, we found that the spindle migration was disturbed after FMNL2 depletion, no matter the cytokinesis occurred or not. Several formin proteins are shown to regulate spindle migration during oocytes meiosis. For example, FMN2 nucleates actin surrounding the spindle, pushing force generated by actin to trigger the spindle migration (19, 22), and cyclin-dependent kinase 1 (Cdk1) induces cytoplasmic Formin-mediated F-actin polymerization to propel the spindle into the cortex (32). Our previous studies also showed that absence of the formin family member FMNL1 or FHOD1 could lead to the decrease of cytoplasmic actin to prevent the spindle migration (20, 27). We speculated that FMNL2 together with other Formin proteins, conservatively regulate actin-mediated spindle migration during oocyte meiosis.

Another important finding is that through the mass spectrometry analysis we found many candidate proteins which were related with ER, and our results indicated that FMNL2 was essential for the maintenance of ER distribution in the cytoplasm. Moreover, the loss of FMNL2 induced ER stress, showing with altered expression of GRP78 and CHOP. Proper distribution of ER is important for the oocyte quality. ER displays a homogeneous distribution pattern throughout the entire ooplasm during development of oocytes and embryos from diabetic mice (33). During the transition of mouse oocytes from MI to MII phase, actin regulates cortical ER aggregation (34). In addition, Formin2 is shown to colocalize with the ER during oocyte meiosis and the ER-associated Formin2 at the spindle periphery is required for MI chromosome migration (6). In our results we showed that FMNL2 associated with INF2 protein in oocytes. INF2 is an ER-associated protein, and the expression of GFP-INF2 which containing DAD/WH2 mutations causes the ER to collapse around the nucleus (35). We concluded that FMNL2 might regulate INF2 for the distribution of ER in cytoplasm of oocytes.

Besides its roles of ER distribution, it is shown that INF2 also affects mitochondrial length and ER-mitochondrial interaction in an actin-dependent manner (35, 36). It is shown that INF2 regulates Drp1 for mitochondrial fission, and INF2-induced actin filaments may drive initial mitochondrial constriction, which allows Drp1-driven secondary constriction (36, 37). In

addition, we also found many candidate proteins which were related with mitochondria from mass spectrometry analysis. During oocyte meiosis, mitochondria gradually accumulated around the spindle after GVBD, and the spindle-peripheral FMN2 and its actin nucleation activity are important for the accumulation of mitochondria in this region (19). Our results found that FMNL2 depletion caused agglutination of mitochondria and altered MMP level in the cytoplasm, indicating its roles on the mitochondria distribution and functions. Another formin protein mDia1 is shown to be necessary to induce the anchoring of mitochondria along the cytoskeletal in mammalian CV-1 cells and *Drosophila* BG2-C2 neuronal cells(38). Moreover, the formin interaction protein Spire1C binds INF2 to promote actin assembly on mitochondrial surfaces, and Spire1C disruption could reduce mitochondrial constriction and division(39). In addition, our result indicated that cofilin expression decreased in FMNL2 depletion oocytes. Cofilin is an actin-depolymerizing factor and its localization at the mitochondrial fission site is crucial for inducing mitochondrial fission and mitophagy (40). Depleting of cofilin resulted in abnormal interconnection and elongation of mitochondria (41). Together with its roles on ER, these data indicated that FMNL2 might associate with INF2 and cofilin for the actin-based organelle distribution during oocyte meiosis.

Collectively, we provide a body of evidence showing that FMNL2 associates with Formin2 and Arp2/3 complex for actin assembly, which further regulates spindle migration and INF2/Cofilin-related organelle dynamics during mouse oocyte maturation.

Data Availability

All data generated or analyzed during this study are included in this published article

Acknowledgements

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Contributions

MHP and SCS designed the study. MHP performed the majority of the experiments. SML, ZNP, MHS, XHL, JQJ, YZ contributed to the reagents and materials. MHP, XHO and SCS analyzed the data. MHP and SCS wrote the manuscript.

Competing interests

There is no conflict of interest to declare.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

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Abbreviations

- FMNL2: Formin-like 2
- FMNLs: Formin-likes
- GV: germinal vesicle
- GVBD: germinal vesicle breakdown
- ATI: anaphase-telophase I
- MI: metaphase I
- MII: metaphase II
- ER: Endoplasmic reticulum

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Editors

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Reviewer #1 (Public Review):

Summary:

The presented study focuses on the role of formin-like 2 (FMNL2) in oocyte meiosis. The authors assessed FMNL2 expression and localization in different meiotic stages and subsequently, by using siRNA, investigated the role of FMNL2 in spindle migration, polar body extrusion, and distribution of mitochondria and endoplasmic reticulum (ER) in mouse oocytes.

Strengths:

Novelty in assessing the role of formin-like 2 in oocyte meiosis

Weaknesses:

Overstating some of the presented data

Unconvincing analysis of the endoplasmic reticulum and mitochondria distribution

The authors addressed all my comments. The section materials and methods was improved. However, some statements still need to be clarified, as they seem to be overstated. I'm still not convinced about the main findings. For example, the analysis of ER and mitochondria distribution was based on a subjective assessment of clustering in meiosis I oocytes, and it's missing objective parameters and timing of the analysis.

Comments on revised version:

The authors addressed all my comments. The section materials and methods was improved. However, some statements still need to be clarified, as they seem to be overstated.

<https://doi.org/10.7554/eLife.92732.2.sa1>

Reviewer #2 (Public Review):

Summary:

This research involves conducting experiments to determine the role of Fmnl2 during oocyte meiosis I.

Strengths:

Identifying the role of Fmnl2 during oocyte meiosis I is significant.

Weaknesses:

The quantitative analysis and the used approach to perturb FMNL2 function would benefit from more confirmatory approaches and rigorous analysis.

Comments on revised version:

The authors addressed most of my comments. However, some comments were not addressed convincingly.

My concern is still valid. The authors used only one approach to knockdown FMNL2 which is "siRNA-mediated knockdown". Using an additional approach to inhibit FMNL2 (Trim-Away or morpholino,...) would be beneficial to confirm that the effect of siRNA-mediated knockdown of FMNL2 is specific.

Response 1: In the author's response, they mentioned that successful migration was quantified based on the contact between the spindle pole and the oocyte cortex. After spindle migration, it is very common for the spindle to be close to (but not in contact with) the cortex for a considerable time. The spindle pole comes in contact with the cortex later (just before anaphase onset and polar body extrusion). Fig. 3A shows an example where at 9 h, the spindle is already migrated but did not come in contact with the cortex until 9:30 h. Based on Fig. 3B,C, the authors assessed spindle migration in fixed oocytes, making it impossible to fix all oocytes at the time of spindle contact with the cortex. Also, the representative images in Fig. 3C do not show spindle staining to assess the contact between the spindle and the cortex.

Overall, I still believe that the distance between the spindle and the cortex is more accurate for quantifying spindle migration.

Response 2: The authors mentioned, "we made appropriate modifications to the relevant descriptions of immunoprecipitation experiments". I can't find these modifications in the manuscript. The authors need to state clearly that the immunoprecipitation results do not necessarily reflect meiotic oocytes specifically because these experiments were done using the whole ovary which contains both somatic cells and oocytes.

Response 5: The authors mentioned that "Based on our observations, during the extrusion of the first polar body in oocytes, there is a temporary occurrence of cellular morphological fragmentation due to cortical reorganization". Unfortunately, this means that the live imaging system in the authors' laboratory is not ideal for oocyte maturation. Several publications show normal oocyte morphology during cytokinesis. Please delete or replace Fig. 2E.

<https://doi.org/10.7554/eLife.92732.2.sa0>

Author response:

The following is the authors' response to the original reviews.

eLife assessment

This study presents useful findings regarding the role of formin-like 2 in mouse oocyte meiosis. The submitted data are supported by incomplete analyses, and in some cases, the conclusions are overstated. If these concerns are addressed, this paper would be of interest to reproductive biologists.

Public Reviews:**Reviewer #1 (Public Review):***Summary:*

The presented study focuses on the role of formin-like 2 (FMNL2) in oocyte meiosis. The authors assessed FMNL2 expression and localization in different meiotic stages and subsequently, by using siRNA, investigated the role of FMNL2 in spindle migration, polar body extrusion, and distribution of mitochondria and endoplasmic reticulum (ER) in mouse oocytes.

Strengths:

Novelty in assessing the role of formin-like 2 in oocyte meiosis.

Weaknesses:

Methods are not properly described.

Overstating presented data.

It is not clear what statistical tests were used.

My main concern is that there are missing important details of how particular experiments and analyses were done. The material and methods section are not written in the way that presented experiments could be repeated - it is missing basic information (e.g., used mouse strain, timepoints of oocytes harvest for particular experiments, used culture media, image acquisition parameters, etc.). Some of the presented data are overstated and incorrectly interpreted. It is not clear to me how the analysis of ER and mitochondria distribution was done, which is an important part of the presented data interpretation. I'm also missing important information about the timing of particular stages of assessed oocytes because the localization of both ER and mitochondria differs at different stages of oocyte meiosis. The data interpretation needs to be justified by proper analysis based on valid parameters, as there is considerable variability in the ER and mitochondria structure and localization across oocytes based on their overall quality and stage.

Thank you for your comment. We regret the oversight of omitting critical information in the manuscript. In the revised manuscript, we have included essential details such as mouse strains, culture media, stages of oocyte and statistical methods in the materials and methods section. Please find our details responses in the “Recommendations for the authors” part.

Reviewer #2 (Public Review):*Summary:*

This research involves conducting experiments to determine the role of Fmnl2 during oocyte meiosis I.

Strengths:

Identifying the role of Fmnl2 during oocyte meiosis I is significant.

Weaknesses:

The quantitative analysis and the used approach to perturb FMNL2 function are currently incomplete and would benefit from more confirmatory approaches and rigorous

analysis.

(1) Most of the results are expected. The new finding here is that FMNL2 regulates cytoplasmic F-actin in mouse oocytes, which is also expected given the role of FMNL2 in other cell types. Given that FMNL2 regulates cytoplasmic F-actin, it is very expected to see all the observed phenotypes. It is already established that F-actin is required for spindle migration to the oocyte cortex, extruding a small polar body and normal organelle distribution and functions.

Thank you for your comment. In the recent decade, Arp2/3 complex (Nat Cell Biol 2011), Formin2 (Nat Cell Biol 2002, Nat Commun 2020), and Spire (Curr Biol 2011) were reported to be 3 key factors to involve into this process. These factors regulate actin filaments in different ways. However, how they cross with each other for the subcellular events were still fully clear. Our current study identified that FMNL2 played a critical role in coordinating these molecules for actin assembly in oocytes. Our findings demonstrate that FMNL2 interacts with both the Arp2/3 complex and Formin2 to facilitate actin-based meiotic spindle migration. Additionally, we discovered a novel role for FMNL2 in determining the distribution and function of the endoplasmic reticulum and mitochondria, which may in turn influence meiotic spindle migration in oocytes. Our results not only uncover the novel functions of FMNL2-mediated actin for organelle distribution, but also extend our understanding of the molecular basis for the unique meiotic spindle migration in oocyte meiosis.

(2) The authors used Fmnl2 cRNA to rescue the effect of siRNA-mediated knockdown of Fmnl2. It is not clear how this works. It is expected that the siRNA will also target the exogenous cRNA construct (which should have the same sequence as endogenous Fmnl2) especially when both of them were injected at the same time. Is this construct mutated to be resistant to the siRNA?

Thank you for your question. We regret any misunderstanding that may have been caused by the inappropriate description in our manuscript. In the rescue experiments, we initially injected FMNL2 siRNA into oocytes, followed by the microinjection of FMNL2 mRNA 18-20 hours later. After conducting our previous experiments, we have verified through Western blotting that endogenous FMNL2 is effectively suppressed 18-20 hours following the microinjection of FMNL2 siRNA. Additionally, we observed a significant increase in exogenous FMNL2 protein expression 2 hours after the injection of FMNL2 mRNA. We believe that the exogenous FMNL2 could compensate the decrease by FMNL2 knockdown, and this approach was adopted in many oocyte studies.

(3) The authors used only one approach to knockdown FMNL2 which is by siRNA. Using an additional approach to inhibit FMNL2 would be beneficial to confirm that the effect of siRNA-mediated knockdown of FMNL2 is specific.

Thank you for your question. Yes, the specificity is always the concern for siRNA or morpholino microinjection due to the off-target issue. Due to the limitation we could not generate the knock out model, and there are no known inhibitors with specific targeting capabilities for FMNL2. To solve this, we performed the rescue study with exogenous mRNA to confirm the effective knock down of FMNL2. These measures provide reassurance regarding the credibility of the experimental outcomes, and this is also the general way to avoid the off-target of siRNA or morpholino.

Reviewer #3 (Public Review):

Summary:

The authors focus on the role of formin-like protein 2 in the mouse oocyte, which could play an important role in actin filament dynamics. The cytoskeleton is known to influence a number of cellular processes from transcription to cytokinesis. The results show that downregulation of FMNL2 affects spindle migration with resulting abnormalities in cytokinesis in oocyte meiosis I.

Weaknesses:

The overall description of methods and figures is overall dismissively poor. The description of the sample types and number of replicate experiments is impossible to interpret throughout, and the quantitative analysis methods are not adequately described. The number of data points presented is unconvincing and unlikely to support the conclusions. On the basis of the data presented, the conclusions appear to be preliminary, overstated, and therefore unconvincing.

Thank you for your comment. We regret the oversight of omitting critical information in the manuscript. In the revised manuscript, we have incorporated your suggestions for modification, particularly regarding the Materials and Methods section. Please see the detailed revision and responses in the “Recommendations for the authors” part.

Recommendations for the authors:

Reviewer #1 (Recommendations for The Authors):

My main concern is that there are missing important details of how particular experiments and analyses were done. The material and methods section is not written in the way that presented experiments could be repeated - it is missing basic information (e.g., used mouse strain, timepoints of oocytes harvest for particular experiments, used culture media, image acquisition parameters, etc.). Some of the presented data are overstated and incorrectly interpreted. It is not clear to me how the analysis of ER and mitochondria distribution was done, which is an important part of the presented data interpretation. I'm also missing important information about the timing of particular stages of assessed oocytes because the localization of both ER and mitochondria differs at different stages of oocyte meiosis. The data interpretation needs to be justified by proper analysis based on valid parameters, as there is considerable variability in the ER and mitochondria structure and localization across oocytes based on their overall quality and stage. My specific comments are listed below.

(1) Information about statistical tests that were used needs to be provided for all quantification experiments.

Thank you for your suggestion. Based on your suggestions, we revised the statistical analysis description in the Materials and Methods section. Additionally, we also included a description of the statistical methods in the legends of the relevant result figures.

(2) I recommend replacing the plunger plots, used in most quantification data, with alternatives allowing evaluation of the distribution of the data (dot plots, box plots, whisker plots).

Thank you for your suggestion. Following your suggestion, we replaced the plunger plots in Fig 2C, D, H, I and Fig3 B, C with dot plots.

(3) Can the authors provide information about particular time points when were individual oocyte stages (GVBD, meiosis I, and meiosis II) harvested/used for immunofluorescence protein detection, western blotting, microinjection, and ER and

mitochondria staining? Were the time points always the same in all presented experiments and experimental vs control group? If not, this needs to be clarified.

Thank you for your suggestion. We used oocytes in the metaphase I (MI) stage for the statistical analysis of spindle migration, actin filament aggregation, endoplasmic reticulum localization, and mitochondrial localization. In the Western blot analysis, GV stage oocytes were utilized to evaluate the efficiency of knockdown and rescue experiments. The protein expression levels of Arp2, Formin2, INF2, Cofilin, Grp78, and Chop in different treatment groups were detected using MI-stage oocytes. In the revised version, we provided all the detailed information about the stages.

(4) Figure 1B: Can the authors comment on why there is a missing representative image of MII oocyte FMNL2-Ab? I recommend including this in the figure to have a complete view of comparing overexpressed and endogenous FMNL2 localization in oocyte meiosis.

Thank you for your suggestion. In the revised manuscript, we added immunostaining images of FMNL2 antibody in MII stage oocytes.

(5) Figure 1C: The figure legend says, "FMNL2 and actin overlapped in cortex and spindle surrounding". In MI oocytes, there is usually no accumulated actin signal around the spindle, which is also true in the presented images, so there cannot be overlapping with the FMNL2 signal. The interpretation should be changed.

We apologize for this inappropriate description that was used, and we deleted this sentence.

(6) Figure 2B: What were the parameters of the "large" and "normal" polar bodies for performing the analysis?

Thank you for your question. In order to assess the size of the polar body, we conducted a comparison between the diameter of the polar body and that of the oocyte. If the diameter of the polar body was found to be less than 1/3 of the oocyte's diameter, we categorized it as normal-sized polar body. Conversely, if the polar body's diameter exceeded 1/3 of the oocyte's diameter, we categorized it as a large polar body. We have included these details in the Results section of the manuscript.

(7) Figure 2F: Can the authors comment on what can be the second band in the rescue group?

Thank you for your question. In the rescue experiment, we microinjected exogenous FMNL2-EGFP mRNA into the oocytes. As a result, compared to endogenous FMNL2, the protein size increased due to the addition of the EGFP tag, approximately 27 kDa. Hence, in the Western blot bands of the rescue group, the upper band represents the expression of exogenous FMNL2-EGFP, while the lower band corresponds to the expression of endogenous FMNL2. We have provided annotations in the revised Figure 2F to clarify this.

(8) Can the authors comment on the variability of PBE between 2C and 2H in the FMNL2-KD groups? In panel C, the PBE in the KD group was 59.5 {plus minus} 2.82%; in panel H, the PBE in the KD group was 48.34 {plus minus} 4.2%, and in the rescue group, the PBE was 62.62 {plus minus} 3.6%. The rescue group has a similar PBE rate as the KD group in panel C. How consistent was the FMNL2 knockdown across individual replicates? Can the authors provide more details on how the rescue experiment was performed?

Thank you for your question. We believe that the difference in PBE observed in Figure 2C and 2H of the FMNL2-KD group was due to the microinjection times and the duration of in vitro

arrest. The results shown in Figure 2C depict the outcome of a single injection of FMNL2 siRNA into GV stage oocytes, followed by 18 hours of in vitro arrest; the results shown in Figure 2H contain a subsequent additional injection of FMNL2-EGFP mRNA with another 2 hours of arrest. The two rounds of microinjection and the extended period of in vitro arrest both affect oocyte maturation rates.

(9). *Figure 2J and K: What groups were compared together? The used statistic needs to be properly described.*

Thank you for your question. The FMNL2-KD, FMNL3-KD, and FMNL2+3-KD groups were all compared to the Control group, therefore, t-test was used for analysis. We have provided explanations in the revised manuscript.

(10) *Figure 4B and C: Can the authors provide representative images without oversaturated actine signal?*

Thank you for your question. For the analysis of oocyte F-actin, the F-actin are divided into cortex actin and cytoplasmic actin. Due to the contrast during imaging, the strong cortex actin signals affected the detection of cytoplasmic actin, therefore, it is necessary to increase the scanning index, which will cause the overexpose the cortex actin signal. This is for the better observation of the cytoplasmic signals.

(11) *Figure 4G + 5H: Can the authors comment on why they used as a housekeeping gene actin instead of tubulin, which was used in the rest of the WB experiments?*

Thank you for your question. In most of the western blot experiments conducted in this study, we used tubulin as a housekeeping gene. However, due to the supply of antibodies by delivery period, we had GAPDH and actin as well for some experiments. These housekeeping genes were all valid for the study.

(12) *Based on what parameters was ER considered normally or abnormally distributed, and what stages of oocytes were assessed?*

Thank you for your question. In this study, we employed oocytes at the MI stage for the analysis of ER localization. In the MI stage, the ER localized around the spindle, which is regarded as the typical localization pattern. The ER displayed a dispersed distribution throughout the cytoplasm or clustered were categorized as aberrant positioning. We included relevant descriptions in the revised version of the manuscript.

(13) *Figure 5H: As a housekeeping gene was used actin - the quantification is labeled as a Grp78 to tubulin ratio.*

Thank you for pointing out the error. This is a label mistake and we corrected it.

(14) *Information about how JC-1 staining was done needs to be provided.*

Thank you for your carefully reading. We included a description of JC1 staining in the Materials and Methods section.

(15). *Line 231-232: "As shown in Figure 4A" - the text doesn't correspond to the figure.*

Thank you for pointing out the error. We revised this mistake in the revised manuscript by correcting "Fig3A" to "Fig4A."

(16) Line 265: there is probably a missing word "Formin2".

Thank you and we corrected the error and made the necessary changes in the revised manuscript.

Reviewer #2 (Recommendations for The Authors):

(1) Quantification and analysis:

- *Fig. 3B: The rate of spindle migration should be quantified based on the distance from the spindle to the cortex. Also, the orientation of the spindle (Z-position) needs to be taken into consideration.*
- *Fig. 5C, D: It is unclear how the rate of ER distribution was calculated.*
- *Western blot: In many experiments (such as Fig. 5H), the bands are saturated which will prevent accurate intensity measurements and quantifications.*

For spindle migration, we specifically focused on spindles exhibiting a distinctive spindle-like shape with clear bipolarity to eliminate any statistical discrepancies potentially caused by variations in Z-axis alignment. Our criterion for determining successful migration was based on the contact between the spindle pole and the cortical region of the oocyte. Therefore, we think that the rate is better to reflect the phenotype than the distance.

For the examination of ER localization, Reviewer 1 also raised this issue. We utilized oocytes at the MI stage in this study. The ER localized around the spindle in MI stage. The ER displayed a dispersed distribution throughout the cytoplasm or clustered were categorized as aberrant positioning. We included relevant descriptions in the revised version of the manuscript.

For the bands of the western blot results, during the experimental procedure we typically capture multiple images at different exposure levels (3-5 images). In the revised manuscript, we have replaced the inappropriate images with more suitable ones.

(2) Given that all Immunoprecipitation experiments in this manuscript were performed on the whole ovary which contains more somatic cells than oocytes, the results do not necessarily reflect meiotic oocytes. Please consider this possibility during the interpretation.

Thank you for your suggestion. Yes, we agree with you. In the revised manuscript, we made appropriate modifications to the relevant descriptions.

(3) 351-365: The conclusion that Arp2/3 compensates for the decreased formin 2 in FMNL2 knockdown oocytes is a bit unconvincing. 1- In mouse oocytes, it is already known that Arp2/3 and formin 2 regulate different pools of F-actin nucleation. 2- The authors found an increase in Arp2/3 in FMNL2 knockdown oocytes compared to control oocytes without any change in cortical F-actin. Given that Arp2/3 is primarily promoting cortical F-actin, it is expected to see an increase in cortical F-actin in FMNL2 knockdown oocytes, which was not the case.

Thank you for your question. Yes, previous studies showed that formin2 localizes to the cytoplasm of oocytes and accumulates around the spindle, which facilitate cytoplasmic actin assembly. While Arp2/3 is primarily responsible for actin assembly at the cortex region of

oocytes. In invasive cells, FMNL2 is mainly localized in the leading edge of the cell, lamellipodia and filopodia tips, to improve cell migration ability by actin-based manner (Curr Biol 2012). We showed that FMNL2 localized both at spindle periphery and cortex, but depletion of FMNL2 did not affect cortex actin intensity. We think that FMNL2 and Arp2/3 both contribute to the cortex actin dynamics, when FMNL2 decreased, ARP2 increased to compensate for this, which maintained the cortex actin level. In the revised manuscript, we have made modifications to avoid excessive extrapolation from our results, ensuring that our conclusions are presented in a more objective manner.

(4) Lines 195-197: The spindle is initially formed soon after the GVBD, so there is no spindle during GVBD. Also, I can't see oocytes at anaphase I or telophase I in this figure. Please revise.

Thank you for your suggestion. We apologize for the inappropriate descriptions that were used. In the revised manuscript, we have made modifications to the respective descriptions in the Results part.

(5) Fig. 2E: It seems that the control oocyte is abnormal with mild cytokinesis defects. Please replace or delete it since this information is already included in Fig. 3A.

Thank you for your suggestion. Based on our observations, during the extrusion of the first polar body in oocytes, there is a temporary occurrence of cellular morphological fragmentation due to cortical reorganization (11h in control oocyte from Fig 2E). However, after the extrusion of the first polar body, the oocyte morphology returns to normal. Figure 2E illustrates the meiotic division process of oocytes, while Figure 3A primarily focuses on the process of oocyte spindle migration. We think that it is better to retain both to present our results.

Reviewer #3 (Recommendations for The Authors):

In the case of the observed phenotype, the stage of GV is important. The phenotypes presented also occur in meiotic or developmentally incompetent oocytes. In addition, the images of GV oocytes appear as NSN, which also show the KD phenotype in Figs. 2 and 3.

Thank you for your concern. As the oocyte grows, the proportion of SN-type oocytes gradually increases. When the oocyte diameter reaches 70-80 μm , the proportion of SN oocytes is approximately 52.7% (Mol Reprod Dev. 1995). In our study, both the control and knockdown groups collected oocytes with a diameter of around 80 μm , which is considered as fully-grown oocytes, predominantly in the SN phase. Since the collection period and size of the oocytes were consistent, we can sure that the observed differences between the control and knockdown groups in phenotype analysis could be solid and reliable.

MII is absent in Fig. 1B.

In the revised manuscript, we added immunostaining images of FMNL2 in MII stage oocytes.

The result of KD is not convincing. Also, discuss whether the heterozygous effect of Fmnl2 deletion affects reproductive fitness.

Thank you for your concern. In our investigation, limited to the setup of knock out model, we employed siRNA to knockdown FMNL2 expression, to avoid the risk of off-target, we performed rescue experiment with exogenous mRNA, which we believe that it could solve this issue. When designing siRNA sequences, we ensured their specificity for binding to FMNL2 mRNA only, and we assessed the levels of FMNL2 and FMNL3 mRNA in oocytes after

injection of FMNL2 siRNA. The results showed that, compared to the control group, the expression of FMNL2 mRNA decreased by approximately 70% after 18 hours of FMNL2 siRNA injection, while the level of FMNL3 mRNA was not decreased.

Fig. 2F rescue experiment with double bands. What bands are seen here? Did the authors inject tagged or untagged FMNL2? Or does endogenous FMNL2 appear higher in the sample after KD?

Thank you for your question. In the rescue experiment, we microinjected exogenous FMNL2-EGFP mRNA into the oocytes. As a result, compared to endogenous FMNL2, the protein size increased due to the addition of the EGFP tag, approximately 27 kDa. Hence, in the Western blot bands of the rescue group, the upper band represents the expression of exogenous FMNL2-EGFP, while the lower band corresponds to the expression of endogenous FMNL2. We provided annotations in the revised Figure 2F to clarify this.

Variability in mitochondria and ER distribution patterns is also known in healthy and developing oocytes, although the authors described only a single phenotype.

Thank you for your concern. Yes, mitochondria and ER show dynamic localization in different stage of oocyte maturation. However, in this study we employed oocyte MI stage for the analysis of ER and mitochondria localization, and in MI stage, both the ER and mitochondria localize around the spindle. This pattern is considered as the normal localization. Several studies showed that dispersed or clustered localization contributed to maturation defects. We included relevant descriptions in the revised manuscript.

What exactly is meant by input in the IP experiments? Why is the target missing in the input sample?

Thank you for your question. We subjected the input samples to electrophoresis on a single channel, all the analyzed proteins demonstrated normal expression, thereby confirming the viability of the input sample. However, upon simultaneous exposure with the IP samples, we observed a lack of clear signal for certain proteins in the input group. This phenomenon is due to the excessive signal intensity resulting from protein enrichment in the IP group, which caused the low exposure of proteins in input group.

Explain the rationale for using, actin or tubulin as loading or normalization controls in the study focusing on the cytoskeleton.

Thank you for your question. Actin and tubulin are both widely used as the control due to their stable expression. For actin, there are α -actin and β -actin isoforms. Formins and Arp2/3 complex regulate the polymerization of α -actin and β -actin to form F-actin, not isoform expression. In our study F-actin (the functional type) was examined. While α -tubulin and β -tubulin are two subtypes of tubulin, and they interact with each other to form stable α/β -tubulin heterodimers. The changes of cytoskeleton dynamics could not change the expression of α/β -tubulin. Therefore, β -actin and α -tubulin could be used as normalization controls.

*Fig. 6E shows only *, but the legend says **.*

Thank you for pointing out the error. We correct the mistake in the revised manuscript.

Spindle positioning appears to differ between control and KD. Does this affect the quantification of Fig. 6F? Adequate nomenclature should be used here.

Thank you for your question. Yes, spindle positioning was affected by FMNL2 depletion. However, central spindle or cortex spindle all belong to MI stage, and JC1 is not related with the stage difference. To avoid misunderstanding we replaced the representative images and corresponding description in Figure 6F.

| *The description of the methods and legends should be significantly improved.*

Thank you for your suggestion. Reviewer 1 and 2 also raised the similar concern. We enriched the description of methods and legends in the revised manuscript.